Project Title

AI-Powered Smart Farming Advisor using IBM watsonx Orchestrate

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Problem Statement

Agriculture is the backbone of many economies, yet farmers often face challenges in selecting suitable crops, managing irrigation, applying fertilizers efficiently, and identifying crop diseases. Due to changing weather conditions, soil variations, and limited access to agricultural experts, farmers may make decisions that reduce crop yield and increase production costs.

To address these challenges, there is a need for an intelligent AI-powered assistant that can provide timely, personalized, and reliable farming recommendations.

Objectives

- Develop an AI-powered Smart Farming Advisor using IBM watsonx Orchestrate.
- Recommend suitable crops based on user queries.
- Provide fertilizer recommendations.
- Suggest irrigation practices.
- Help identify common crop diseases and pests.
- Promote sustainable and environmentally friendly farming.
- Use a knowledge base to provide reliable agricultural guidance.

Proposed Solution

The Smart Farming Advisor is an AI-powered conversational assistant built using IBM watsonx Orchestrate. The agent interacts with farmers through natural language and provides personalized recommendations using a curated agricultural knowledge base.

The knowledge base contains information on crop selection, fertilizers, irrigation practices, pest and disease management, and sustainable farming methods. The agent retrieves relevant information and generates practical recommendations for farmers.

IBM Technologies Used

- IBM Cloud

- IBM watsonx Orchestrate
- IBM Foundation Model (Granite / GPT Foundation Model)
- AI Agent Builder
- Knowledge Base
- PDF Knowledge Sources

Expected Outcome

The Smart Farming Advisor will:

- Recommend suitable crops.
- Suggest fertilizers.
- Provide irrigation guidance.
- Help identify crop diseases.
- Promote sustainable farming.
- Improve decision-making for farmers.

Conclusion

The Smart Farming Advisor demonstrates how IBM watsonx Orchestrate can be used to build intelligent AI assistants for agriculture. By combining conversational AI with a curated agricultural knowledge base, the system provides practical recommendations that support better farming practices and sustainable agriculture.